

# PASCA 2.0

Summer School at CIMAT



## CIMAT

Guanajuato, Mexico

## Lecture Notes

# Singularities of Foliations in Characteristic $p$

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# 1 An Invitation to $p$ -Singularities

We begin with some motivating questions about foliations, differential equations, and singularities in characteristic  $p$ .

**Question 1.1.** What does it mean for a PDE, or more geometrically for a singular space governed by differential equations, to be  $F$ -rational?

**Question 1.2.** What is a foliation?

## Geometric Motivation

**Example 1.3** (Elementary pictures of foliations). Consider a small open interval or disk, for instance

$$X = (-\epsilon, \epsilon)$$

(1) The equation

$$x^2 + y^2 = c$$

(2) The equation

$$\frac{y}{x} = c$$

These two pictures represent two basic ways in which a space may be decomposed into lower-dimensional leaves.

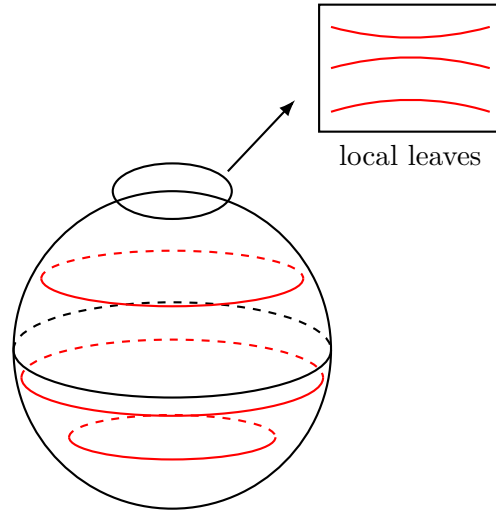
**Remark 1.4** (Added Context). A foliation is a decomposition of a space into subspaces called leaves, which locally look like parallel slices. In differential geometry, these leaves are usually integral manifolds of a distribution, i.e. of a subbundle of the tangent bundle.

## Foliations on Manifolds

Let  $X$  be a smooth manifold and let  $T_X$  denote its tangent bundle. A distribution is a subbundle

$$F \subseteq T_X.$$

**Theorem 1.5** (Frobenius Theorem). *Let  $X$  be a smooth manifold and let  $F$  be a subbundle. Then  $F$  is integrable if and only if the following conditions hold:*



**Figure 1.** A schematic foliation by leaves on a surface, with a local chart.

(1)  $F$  is closed under the Lie bracket of vector fields, i.e.

$$[F, F] \subseteq F;$$

Thus one has the following dictionary:

$$\left\{ \text{foliations } \mathcal{F} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{integrable subbundles} \\ T_{\mathcal{F}} = D \subseteq T_X \end{array} \right\}.$$

**Example 1.6** (Concentric-circle foliation). The foliation defined by the level sets

$$x^2 + y^2 = c$$

yields:

$$d(x^2 + y^2) = 0.$$

$$2x \, dx + 2y \, dy = 0,$$

$$x \, dx + y \, dy = 0.$$

$$\mathcal{F} = \int D.$$

## Foliations in Commutative Algebra

We now translate the differential-geometric notion into commutative algebra.

Let  $k$  be a field and let  $A$  be a  $k$ -algebra which is a normal integral domain. Let

$$\mathrm{Der}_k(A)$$

denote the module of derivations.

**Definition 1.7** (Algebraic foliation). A *foliation* on  $A$  is an  $A$ -submodule

$$\mathcal{F} \subseteq \mathrm{Der}_k(A)$$

satisfying:

(1) **Saturation:** the quotient

$$\mathrm{Der}_k(A)/\mathcal{F}$$

is torsion-free.

(2) **Lie-closedness:**  $\mathcal{F}$  is closed under the Lie bracket:

$$[\mathcal{F}, \mathcal{F}] \subseteq \mathcal{F}.$$

**Remark 1.8.** The saturation condition is the algebraic analogue of requiring the quotient to behave like a vector bundle on a dense open set. Since  $A$  is normal, torsion-freeness is a natural replacement for local freeness in singular settings.

## The Characteristic $p$ Miracle

Assume now that  $k$  has characteristic  $p > 0$ . In characteristic  $p$ , foliations are closely related to intermediate rings between  $A^p$  and  $A$ .

### 1.1 The Ekedahl–Jacobson correspondence

**Theorem 1.9** (Ekedahl–Jacobson Correspondence). *Let  $A$  be a normal domain over a field  $k$  of characteristic  $p$ . There is a one-to-one correspondence*

$$\left\{ \begin{array}{l} R \text{ normal such that} \\ A^p \subseteq R \subseteq A \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \mathcal{F} \subseteq \mathrm{Der}_k(A) \\ \mathcal{F} \text{ a foliation} \end{array} \right\}.$$

given by:

$$\begin{aligned} R &\longmapsto \mathrm{Der}_R(A) \\ \mathcal{F} &\longmapsto A^{\mathcal{F}}, \end{aligned}$$

where

$$A^{\mathcal{F}} = \{a \in A \mid \partial(a) = 0 \text{ for all } \partial \in \mathcal{F}\}.$$

**Remark 1.10.** The containment  $A^p \subseteq A^{\mathcal{F}}$  holds because every derivation kills  $p$ -th powers in characteristic  $p$ :

$$\partial(a^p) = pa^{p-1}\partial(a) = 0.$$

$$\left\{ \begin{array}{l} A^p \subseteq R \subseteq A \\ R \text{ normal} \end{array} \right\} \begin{array}{c} \xrightarrow{R \mapsto \text{Der}_R(A)} \\ \xleftarrow{\mathcal{F} \mapsto A^{\mathcal{F}}} \end{array} \left\{ \begin{array}{l} \mathcal{F} \subseteq \text{Der}_k(A) \\ \mathcal{F} \text{ foliation} \end{array} \right\}$$

## A foliated Kunz theorem

**Theorem 1.11** (Foliated Kunz Theorem). *Let  $A$  be a smooth  $k$ -algebra of characteristic  $p$ . Under the Ekedahl–Jacobson correspondence*

$$A^{\mathcal{F}} \longleftrightarrow \mathcal{F},$$

*the following are equivalent:*

- (1)  $A^{\mathcal{F}}$ , equivalently  $A$ , is smooth;
- (2)  $A$  is flat over  $A^{\mathcal{F}}$ ;
- (3)  $A$  is locally free, equivalently flat, over  $A^{\mathcal{F}}$ .

**Remark 1.12.** This statement is a foliated analogue of Kunz’s theorem. Classical Kunz says that a Noetherian ring of characteristic  $p$  is regular if and only if the Frobenius morphism is flat. Here, flatness of the quotient by the foliation detects smoothness of the ring of constants.

## Examples

**Example 1.13** (Euler derivation on the affine plane). Let

$$A = k[x, y]$$

and consider the Euler derivation:

$$\partial = x\partial_x + y\partial_y.$$

We define the foliation:

$$\mathcal{F} = \langle x\partial_x + y\partial_y \rangle_A.$$

Notice that:

$$\partial(x^a y^b) = (a + b)x^a y^b.$$

This is zero if and only if:

$$a + b \equiv 0 \pmod{p}.$$

Thus, the constants include:

$$x^p, \quad x^{p-1}y, \quad x^{p-2}y^2, \quad \dots, \quad xy^{p-1}, \quad y^p$$

giving:

$$A^{\mathcal{F}} = k[x^p, x^{p-1}y, x^{p-2}y^2, \dots, xy^{p-1}, y^p].$$

**Remark 1.14.** More precisely, the invariant ring is generated by all monomials  $x^a y^b$  with  $a + b \equiv 0 \pmod{p}$ . The displayed generators give the degree- $p$  Veronese subring; depending on the precise foliation and saturation convention, one should verify whether the full constant ring is exactly this Veronese subring or its integral closure inside  $A$ . In the standard Euler-derivation example, the  $p$ -th Veronese description captures the intended lecture-note picture.

## $F$ -Rationality and Differential Operators

We now recall the general philosophy behind  $p$ -singularities.

Let  $A$  be a local ring of characteristic  $p$ . Roughly speaking,  $A$  is  $F$ -rational if it is Cohen–Macaulay and parameter ideals are tightly closed. Equivalently, for excellent local rings,  $F$ -rationality can be detected by the behavior of Frobenius actions on local cohomology or by trace maps on canonical modules.

**Remark 1.15.** We have a connection between foliations, differential operators, and  $F$ -rationality. The exact formulation depends on the chosen singularity class:  $F$ -rational,  $F$ -regular, or  $F$ -pure. The common theme is that Frobenius-linear maps and Cartier operators control the singularities.

Suppose that  $A$  is finite over  $A^p$ . For instance, this occurs when  $A$  is  $F$ -finite. Let  $\omega_A$  be a canonical module. In the regular or finite setting one may write

$$\omega_A \cong \operatorname{Hom}_{A^p}(A, A^p),$$

For  $F$ -singularities, one studies maps induced by Frobenius trace:

$$F_*^e \omega_A \longrightarrow \omega_A.$$

Or equivalently:

$$\begin{aligned} F_*^e \omega_A &\xrightarrow{F_*^e c} F_*^e \omega_A \longrightarrow \omega_A. \\ F_*^e \omega_A &\xrightarrow{F \cdot c} F_*^e \omega_A \longrightarrow \frac{\omega_A}{(\mathrm{Sp})}. \end{aligned}$$

### Differential operators in characteristic $p$

The ring of  $k$ -linear differential operators on  $A$  can be described by

$$D_A = \bigcup_{e \geq 0} \mathrm{Hom}_{A^{p^e}}(A, A) \subseteq \mathrm{Hom}_k(A, A).$$

We denote the level- $e$  differential operators as:

$$D_A^{(e)} := \mathrm{Hom}_{A^{p^e}}(A, A)$$

A foliation  $\mathcal{F}$  may be viewed as an  $A$ -submodule, or more generally as part of a filtered structure inside the ring of differential operators. We denote its level- $e$  component by

$$\mathcal{F}^{(e)} = \mathcal{F} \cap D_A^{(e)}.$$

The corresponding rings of constants form a descending chain

$$A \supseteq A^{(1)} \supseteq A^{(2)} \supseteq \dots \supseteq A^{(e)} \supseteq \dots,$$

where:

$$A^{(e)} = A^{\mathcal{F}^{(e)}}.$$

Explicitly:

$$A^{(e)} = \{a \in A \mid [\delta, a] = 0 \text{ for all } \delta \in \mathcal{F}^{(e)}\}.$$

**Remark 1.16.** The commutator notation  $[\delta, a]$  views  $a$  as the operator “multiplication by  $a$ .” Thus  $[\delta, a] = 0$  means that the operator  $\delta$  commutes with multiplication by  $a$ . This is a natural higher-level analogue of saying that a derivation kills  $a$ .

## Divided Powers and Higher Constants

**Example 1.17** (Higher constants for the Euler foliation). Let

$$A = k[x, y]$$

and:

$$\partial = x\partial_x + y\partial_y.$$

We can form the module of divided powers:

$$M = \bigoplus_{k \geq 0} A \partial^{(k)},$$

where:

$$\partial^{(k)} = \sum_{i+j=k} x^i y^j \partial_x^{(i)} \partial_y^{(j)}.$$

The  $e$ -th constant ring is described by

$$A^{(e)} = k[x^{p^e}, x^{p^e-1}y, x^{p^e-2}y^2, \dots, xy^{p^e-1}, y^{p^e}].$$

With:

$$K_0 := \text{Der}(A^{(e)})$$

**Remark 1.18.** The appearance of the generators

$$x^{p^e}, x^{p^e-1}y, \dots, y^{p^e}$$

reflects the higher-order constants under the divided power derivations.

## Cartier Foliations

**Definition 1.19** ( $F$ -foliation or Cartier foliation). An  $F$ -foliation, also called a *Cartier foliation*, is a compatible higher-level foliation for which the corresponding system of constant rings behaves stably in the sense that

$$A^{(e)} = A^{(e+1)} \quad \text{for } e \gg 0,$$

**Remark 1.20.** In many treatments, a Cartier structure is not literally defined by equality  $A^{(e)} = A^{(e+1)}$  for large  $e$ , but rather by compatible Frobenius-linear or Cartier-linear maps at all levels. The stabilization condition above records the intended lecture-note



heuristic: an  $F$ -foliation is a foliation equipped with coherent data through all Frobenius levels.

In the local setting, let  $A$  be a regular local ring. One studies  $\mathcal{F}$  as an  $F$ -regular Cartier foliation of rank  $r$ . Such a foliation is locally governed by variables

$$x_1, \dots, x_r$$

**Definition 1.21** (Partial  $p$ -basis). Let  $A$  be an  $F$ -finite ring of characteristic  $p$ . A collection

$$x_1, \dots, x_r \in A$$

forms a partial  $p$ -basis if they extend to a  $p$ -basis of  $A$  over  $A^p$ .

**Theorem 1.22** (Cartier operators attached to a Cartier foliation). *For a Cartier foliation  $\mathcal{F}$ , there is a compatible system of Cartier operators*

$$\kappa_e : F_*^e \omega_{\mathcal{F}_e} \longrightarrow \omega_{\mathcal{F}_e},$$

**Remark 1.23.** The operators  $\kappa_e$  are the foliated analogue of Frobenius trace maps. Properties such as  $F$ -purity, sharp  $F$ -purity,  $F$ -regularity, and  $F$ -rationality are often formulated in terms of the surjectivity, splitting, or generation behavior of such Cartier operators.

## Summary of the Main Dictionary

The lecture develops the following chain of ideas:

$$\text{foliations} \quad \longleftrightarrow \quad \text{Lie-closed saturated submodules of } \text{Der}_k(A)$$

$$\text{foliations} \quad \longleftrightarrow \quad \text{intermediate rings } A^p \subseteq R \subseteq A.$$

Using the ring of differential operators:

$$D_A = \bigcup_{e \geq 0} \text{Hom}_{A^{p^e}}(A, A),$$

we get the following coherent picture:

$$\begin{array}{ccc}
\text{foliation } \mathcal{F} & \xrightarrow{\text{constants}} & A^{\mathcal{F}} \\
\text{higher levels} \downarrow & & \downarrow p^e\text{-level constants} \\
\mathcal{F}^{(e)} \subseteq D_A^{(e)} & \xrightarrow{\text{constants}} & A^{(e)}
\end{array}$$

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